



CALICUT UNIVERSITY – FOUR-YEAR UNDER GRADUATE PROGRAMME (CU-FYUGP)

BSc PHYSICS HONOURS

Programme	B.Sc. Physics Honours				
Course Title	DATA ANALYSIS IN PHYSICS USING MACHINE LEARNING				
Type of Course	Vocational Minor (GROUP II: DATA ANALYSIS IN PHYSICS)				
Semester	III				
Academic Level	200 - 299				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	3	-	2	75
Pre-requisites	1. Fundamentals of Programming Concepts 2. PHY1VN102- Python Basics 3. PHY2VN102- Data Analysis in Physics Using Python				
Course Summary	This course explores Machine Learning fundamentals: types, challenges, and model training techniques like Linear Regression, Gradient Descent, KNN, and clustering. Analyze data using Scikit-learn, handle classification problems with performance evaluation measures on real datasets.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Grasp the concepts and importance of Machine Learning, its types, and	U	C	Instructor-created exams / Quiz

	real-world problem-solving applications.			
CO2	Understand linear regression, model evaluation metrics, and various types of regression. They will apply this knowledge practically using examples.	Ap	P	Practical Assignment / Observation of Practical Skills
CO3	Master in K-Nearest Neighbor classification, decision trees, entropy, Gini index, and K-means clustering, demonstrated through practical applications with sample datasets.	Ap	P	Seminar Presentation / Group Tutorial Work
CO4	Apply classification algorithms to MNIST data, including binary classifiers and multilabel classification, and interpret performance measures like confusion matrix, precision, recall, and ROC curve	U	C	Instructor-created exams / Home Assignments
CO5	Learn to implement and construct a ML model for one of the problems mentioned.	Ap	P	One Minute Reflection Writing assignments/ Vice Voce
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs (45 +30)	Marks (70)
I	Foundations of Machine Learning		11	15
	1	Introduction to Machine Learning - Need for Machine Learning - Machine Learning model	1	
	2	Challenges in ML - Applications of ML	1	
	3	Types of ML algorithms - Supervised ML Algorithms - Classification - Regression -	1	

	4	Exploring Unsupervised Learning, Reinforcement Learning -	2	
	5	Preparing Data - Steps involved in data cleaning - Data Standardization - Data Scaling, Binarization - Data Labeling,	3	
	6	Feature Selection Techniques - Detecting Outliers - Z score - Optimization Algorithm - Gradient Descent - SGD	3	
	Sections from 9.1 - 9.9 of Chapter 9 of Book 2			
II	Regression Analysis: Techniques, Evaluation, and Practical Applications		11	18
	7	Overview of how Regression works - Model evaluation metrics - Types of Regression	2	
	8	Understanding Linear Regression, Simple Linear regression - Variables - Linear Regression - Linear equation - The r-squared value	3	
	9	Practical use of Simple Linear regression - An example problem using sample data (home prices)	3	
	10	Make the data - identify the features - Training and Testing - another example problem for linear regression (Salary data)	1	
	11	Multiple linear regression - Example problem using sample data	2	
	1. Section 10.1 - 10.4 of Chapter 10 of Book 2 2. Chapter 19 page no. 382 - 400 of Book 3 3. Chapter 20 page no. 401 - 408 of Book 3			
III	ML Classification & Clustering Essentials		14	25
	12	Classification Algorithms - K-Nearest Neighbour classifier - How to select K value	2	
	13	Calculate the distance metric between two points - Example problem to construct the classifier - use breast cancer data set	3	
	14	Decision Trees - Entropy - How to calculate total entropy for a dataset	3	
	15	Gini Index	1	
	16	Comparison between Gini index and entropy- Example problem using a given data set	2	
	17	Clustering Algorithms - K- means clustering	1	
	18	Rules to generate clusters - Elbow method - Sample problem using a standard data set	2	
	1. Sections and references from Chapters 29 page no. 572 - 585 of Book 3			

	2. Sections and references from Chapters 30 page no. 591 - 607 of Book 3		
	3. Chapter 11 Section 11.3 - 11.4		
IV	Classification: Metrics & Multilabel Analysis	9	12
	19 Classification problem using MNIST data	2	
	20 Training a binary classifier	2	
	21 Performance Measures - Confusion Matrix - Precision and Recall - ROC curve	3	
	22 Multilabel Classification, multi output classification	2	
	1. Sections from Chapter 3 page no. 85 - 108 of Book 1		
V	Hands-on Data Structures: Practical/Project Applications, Case Study and Course Project	30	
	1 Implement the following: 1. Classification of iris data using KNN: Data: Read from Scikit-learn 2. Classification of iris data using K-means Cluster: Data: Read from Scikit-learn 3. Draw the confusion matrix of iris dat: Data: Use the classification results from experiments 1 & 2 4. Design ML Classifier: To classify RR Lyrae stars using KNN.		
	1. https://scikit-learn.org/stable/auto_examples/neighbors/plot_classification.html#sphx-glr-auto-examples-neighbors-plot-classification-py 2. https://www.geeksforgeeks.org/analyzing-decision-tree-and-k-means-clustering-using-iris-dataset/ 3. https://www.kaggle.com/code/ankumagawa/knn-confusion-matrix-iris-flower-digits-data 4. https://sigmoidal.ai/en/k-nearest-neighbors-k-nn-for-classifying-rr-lyrae-stars/		
Books and References: - Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow: Concepts, Tools, and Techniques to Build Intelligent Systems, Third Edition by Aurélien Géron. (Book 1) - Data Science and Machine Learning using Python by Reema Thereja (Book 2) - Machine Learning in Data Science using Python by R Nageswara Rao (Book 3)			

Mapping of COs with PSOs and POs :

	PSO 1	PSO 2	PSO 3	PSO4	PS O5	PS O6	PO1	PO2	PO3	PO4	PO5	PO 6	PO 7
CO 1	0	0	1	0	1	0	2	0	0	2	1	0	0
CO 2	0	1	2	0	1	0	2	0	0	2	1	0	0
CO 3	0	1	2	0	1	0	2	0	0	2	1	0	0
CO 4	0	1	2	0	1	0	2	0	0	2	2	0	0
CO 5	0	2	1	1	1	0	2	0	1	2	1	0	0
CO 6	0	0	1	0	1	0	2	0	0	2	1	0	0

Correlation Levels:

Level	Correlation
0	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Assignment/ Quiz/ Discussion / Seminar
- Midterm Exam
- Programming Assignments (20%)
- Final Exam (70%)

Mapping of COs to Assessment Rubrics :

	Internal Exam	Assignment	Project Evaluation	End Semester Examinations
CO 1	✓			✓
CO 2	✓			✓
CO 3	✓			✓
CO 4		✓		✓
CO 5		✓		✓
CO 6			✓	